



CLASS PARTICIPATION COVERSHEET

Unit of Study: **MIT653 Research Methods**

Assignment Title: **Class Participation – Week 04**

Semester and Year: **Trimester 2, 2026**

Lecturer Name: **Dr. Sudath Heiyanthuduwage**

DECLARATION

The work contained in this assignment is my own work and has not been copied from other sources or been previously submitted, unless otherwise specifically acknowledged.

I understand that failure to comply can lead to severe penalties.

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Task 1: Keywords for Literature Search

The following keywords and keyword combinations were used to search for relevant academic literature on Artificial Intelligence, Machine Learning, and cybersecurity:

- Artificial intelligence AND cybersecurity
- Machine learning AND intrusion detection
- Deep learning AND cyber attack detection
- Zero-day attack detection
- Malware detection AND deep learning
- Phishing detection AND machine learning
- Adversarial machine learning AND cybersecurity
- Reinforcement learning AND cyber defense
- ChatGPT OR generative AI AND cybersecurity threat
- Network intrusion detection system survey

Task 2: Literature Sources, Academic Databases and Inclusion Criteria

2.1 Types of Literature Sources

- Peer-reviewed journal articles
- Peer-reviewed conference papers
- Systematic review and survey papers (secondary sources used to identify well-established findings and research gaps)

2.2 Academic Databases Searched

- IEEE Xplore Digital Library
- ACM Digital Library
- ScienceDirect
- Springer Digital Library
- MDPI (Sensors journal)
- Google Scholar (used to cross-check citation counts and locate full text)

2.3 Inclusion Criteria

- Published between 2020 and 2024, to ensure the review reflects current AI/ML techniques and threats.
- Directly related to the application of AI, machine learning, or deep learning in a cybersecurity context, such as intrusion detection, malware detection, phishing detection, or adversarial robustness.
- Published in a peer-reviewed journal or reputable conference proceedings.
- Available in full text in English.
- Cited by other academic work or published in an indexed, reputable outlet such as IEEE, ACM, Elsevier, MDPI, Springer, or Wiley.

2.4 Exclusion Criteria

- Non-peer-reviewed blog posts, magazine articles, or vendor marketing material.
- Papers not available in full text.

- Papers unrelated to the cybersecurity domain.

Task 3: Selected Articles and Detailed Analysis

Ten articles were selected because they provide coverage of multiple cybersecurity domains, including intrusion detection, malware detection, phishing detection, adversarial robustness, and generative AI. The selected studies collectively provide evidence relevant to the research question and hypothesis.

#	Citation	Focus Area
1	Ozkan-Okay et al. (2024)	ML, DL, RL and ChatGPT in cybersecurity (general survey)
2	Momand, Jan & Ramzan (2023)	Intrusion detection systems using ML/DL
3	Pinto et al. (2023)	ML-based IDS for critical infrastructure
4	Ali et al. (2024)	ML strategies in intrusion detection systems
5	Bensaoud, Kalita & Bensaoud (2024)	Deep learning for malware detection
6	Deldar & Abadi (2023)	Deep learning for zero-day malware detection
7	Zieni, Massari & Calzarossa (2023)	Phishing website detection survey
8	Safi & Singh (2023)	Phishing website detection techniques (SLR)
9	Macas, Wu & Fuertes (2023)	Adversarial attacks/defenses in DL-based cybersecurity
10	Chowdhury et al. (2023)	ChatGPT as a threat to the CIA triad

3.1 Common Themes Across the Papers

- Traditional, rule-based, or signature-based security systems are consistently reported as too slow and too rigid to keep up with new and evolving cyberattacks, especially zero-day attacks.
- Machine learning and deep learning models such as CNN, LSTM, Random Forest, and SVM are repeatedly shown to outperform traditional detection methods in accuracy across several domains, including network intrusion, malware, and phishing detection.
- Almost every paper raises the limitation that AI/ML-based defences remain vulnerable to adversarial manipulation and can themselves be exploited. Generative AI tools such as ChatGPT can also be used to support attack activities.
- The quality, size, and diversity of training datasets is repeatedly cited as a major factor limiting the realworld performance of AI-based cybersecurity systems.

3.2 Key Findings and Methodologies Used

- Ozkan-Okay et al. (2024) conducted a broad survey comparing supervised, unsupervised, deep learning, and reinforcement learning methods across cybersecurity tasks. The study concludes that no single technique is

superior across all attack types and that combining models through hybrid approaches can increase detection rates.

- Momand, Jan & Ramzan (2023) and Pinto et al. (2023) systematically reviewed ML/DL-based intrusion detection systems. Their work evaluated approaches against benchmark datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15 and organised different cybersecurity attack types.
- Bensaoud, Kalita & Bensaoud (2024) and Deldar & Abadi (2023) focused on deep learning for malware detection. Both indicate that deep learning models can generalise better to previously unseen or zero-day malware than static, signature-based antivirus methods, although they require larger training datasets.
- Zieni, Massari & Calzarossa (2023) and Safi & Singh (2023) reviewed machine-learning-based phishing detection. They report high accuracy for supervised classifiers such as Random Forest and SVM when trained using URL and webpage features, while also identifying rapid concept drift as a challenge.
- Macas, Wu & Fuertes (2023) evaluated adversarial examples that can fool deep-learning-based cybersecurity systems and catalogued defence strategies such as adversarial training.
- Chowdhury et al. (2023) considered the dual-use nature of ChatGPT-like tools, examining both their potential to support cybersecurity operations and their potential to threaten confidentiality, integrity, and availability.

3.3 Research Gaps and Unresolved Issues

- Most reviewed systems are tested on fixed benchmark datasets such as NSL-KDD and CICIDS2017, which do not fully reflect real-time, real-world network traffic. This limits confidence in reported accuracy figures.
- Adversarial robustness remains an unresolved problem because AI-based systems can be fooled by attackers who have some knowledge of the model.
- Many studies report accuracy or efficiency improvements for one specific attack type or domain, such as malware, phishing, or network intrusion, rather than evaluating AI/ML's broader impact on threat detection efficiency and effectiveness across a security system.
- The dual-use risk of generative AI tools, where the same technology can be useful for defence but also exploitable for attack, is an emerging and still under-researched area.

3.4 Contribution of Each Paper to the Research Topic and Research Question

- The general survey by Ozkan-Okay et al. (2024) establishes the broad landscape and supports the hypothesis by showing the adoption of ML, DL, and RL methods in response to the limitations of traditional rule-based approaches.
- The IDS-focused studies by Momand et al. (2023), Pinto et al. (2023), and Ali et al. (2024) provide evidence and evaluation perspectives for the efficiency and effectiveness dimensions of the research question.
- The malware and phishing detection studies provide comparative evidence from additional cybersecurity domains, reinforcing the expectation that ML/DL can improve detection efficiency and accuracy across different threat types.
- The adversarial robustness and ChatGPT studies add balance by identifying limitations and risks. They show that AI/ML efficiency gains are not unconditional and may be reduced by adversarial manipulation or dual-use misuse.

Task 4: Literature Review Outline / Structure

1. Introduction

2. Purpose and Aims of the Literature Review
3. Literature Review Process (keywords, databases, inclusion/exclusion criteria)
4. Thematic Analysis
 - 4.1 Machine Learning and Deep Learning in Intrusion Detection
 - 4.2 Malware Detection Using Deep Learning
 - 4.3 Phishing Detection Using Machine Learning
 - 4.4 Adversarial Robustness and Limitations of AI-Based Defences
 - 4.5 Dual-Use Risk: Generative AI in Cybersecurity
5. Research Gaps Identified
6. Conclusion
7. References

Task 5: Literature Review Report Sections

5.1 Introduction

Cybersecurity threats have grown rapidly in both volume and sophistication, and traditional signature-based detection methods are increasingly unable to identify new or previously unseen (zero-day) attacks. In response, researchers and organisations have turned to artificial intelligence (AI), machine learning (ML), and deep learning (DL) techniques to build more adaptive and accurate threat detection systems. This literature review examines recent academic research on the application of AI/ML techniques across several cybersecurity domains, including intrusion detection, malware detection, and phishing detection, in order to establish the current state of knowledge and identify where further research, such as this study, can contribute.

5.2 Purpose and Aims of the Literature Review

- Establish what is already known about the effectiveness of AI/ML techniques in detecting cyberattacks, particularly zero-day attacks, compared to traditional methods.
- Identify the datasets, models, and evaluation methods commonly used in this field, to inform the design of this research.
- Identify unresolved research gaps that justify and shape the research question and hypothesis of this study.
- Provide a balanced view that considers both the benefits and the limitations and risks of AI-based cybersecurity systems.

5.3 Literature Review Process

A structured search was conducted across IEEE Xplore, ACM Digital Library, ScienceDirect, Springer, and MDPI using the keyword combinations listed in Task 1. Search results were filtered using the inclusion and exclusion criteria described in Task 2, including publication between 2020 and 2024, peer review, direct relevance to AI/ML in cybersecurity, and availability of full text in English. From the initial search results, ten articles were selected based on relevance to the research question, methodological rigour, and coverage of different cybersecurity sub-domains. Each article was read in full, and key findings, methodologies, and limitations were extracted and compared.

5.4 References

- Ali, A. H., Charfeddine, M., Ammar, B., Ben Hamed, B., Albalwy, F., Alqarafi, A., & Hussain, A. (2024). Unveiling machine learning strategies and considerations in intrusion detection systems: A comprehensive survey. *Frontiers in Computer Science*, 6, 1387354. <https://doi.org/10.3389/fcomp.2024.1387354>
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- Momand, A., Jan, S. U., & Ramzan, N. (2023). A systematic and comprehensive survey of recent advances in intrusion detection systems using machine learning: Deep learning, datasets, and attack taxonomy. *Journal of Sensors*, 2023, Article 6048087.
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- Pinto, A., Herrera, L.-C., Donoso, Y., & Gutierrez, J. A. (2023). Survey on intrusion detection systems based on machine learning techniques for the protection of critical infrastructure. *Sensors*, 23(5), 2415.
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☐ **A Comparative Study of Diverse Intrusion Detection Methods using Machine Learning Techniques** ⓘ

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